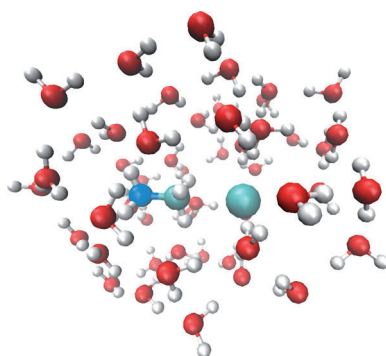


ARTICLES

Waterworld: The influence of water solvent on the Menshutkin reaction of methyl chloride with ammonia is investigated by using static, quantum chemical calculations. Large, explicit, and globally structure-optimized water clusters are employed around the reaction center in a mixed explicit/implicit solvent model. Certain water clusters can completely annihilate the sizeable activation barrier of $23.5 \text{ kcal mol}^{-1}$.



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■■ – ■■

**Assessing Solvation Effects on
Chemical Reactions with Globally
Optimized Solvent Clusters**

Assessing Solvation Effects on Chemical Reactions with Globally Optimized Solvent Clusters

Yan Li and Bernd Hartke^{*[a]}

We have examined the influence of water solvent on the Menschutkin reaction of methyl chloride with ammonia by performing static, quantum chemical calculations. We have employed large, explicit, and globally structure-optimized water clusters around the reaction center, in a mixed explicit/implicit solvent model. This approach deliberately deviates from attempts to capture the most likely solvent-molecule distribution around a reaction center. Instead, it explores extremes on the scale of rearrangement speed in terms of the surrounding solvent cluster,

relative to the reaction progress itself. A comparison to traditional theoretical and experimental results enables us to quantify the energy penalty that is induced by the inability of the water cluster to instantaneously and completely follow the reaction progress. In addition, the influence of water clusters on the reaction energy profile can be much larger than merely changing it somewhat. Certain clusters can completely annihilate the sizeable activation barrier of 23.5 kcal mol⁻¹.

1. Introduction

Solvent molecules play an important role in determining the rate constant and the reaction mechanism in chemical processes that occur in solution.^[1] There is a vast amount of literature data on modeling reactions in solution; as such, we cannot give a balanced overview here. Instead, we cite a few characteristic works and face the risk of leaving out what many readers may view as more important.

Various possible ways to treat solvent molecules have been suggested over the last few years. One of them is the explicit model, which maintains the molecular nature of the solvent. For exemplary work, see refs. [2–6]. This model is by far the most widely employed approach for describing a solvent in terms of molecular dynamics (MD) or Monte Carlo (MC) simulations.^[7–9] If just a small number of solvent molecules are considered, this approach is referred to as supermolecule model.

Another approach is the implicit solvent model. In this approach, the molecular nature of the solvent is ignored. Instead, the solvent is represented by using a polarizable continuum medium with the dielectric constant (ϵ). For exemplary works, see refs. [10–14]; for reviews, see refs. [15, 16].

Both models have advantages and disadvantages. For example, although the explicit model can provide a detailed and accurate description of the solvated system, it is very time-consuming if a sufficient number of solvent molecules are included. In turn, the supermolecule model cannot include long-range electrostatic interactions with the bulk solvent, nor the embedding of a few explicitly treated solvent molecules into their solvent surroundings. On the other hand, the implicit

model cannot provide detailed solvation structures; hence, it cannot treat the specific interactions between the solute and the solvent, such as hydrogen bonding, hydrophobic effects, and so forth.

Hence, the so-called mixed explicit/implicit solvent model, which can take into account both short-range effects and long-range electrostatic interactions (owing to the bulk solvent), is desirable and has been employed by many groups.^[17–27] However, in many of these studies, as discussed in a review on this topic,^[28] the unanimous and explicit conclusion was that the number of solvent molecules that were explicitly included was too small, that the number of possible spatial arrangements was far too large, or both, even for very small numbers of explicit solvent molecules (typically between one and four). Hence, it is desirable to extend this approach to larger numbers of explicitly treated solvent molecules. However, as was also explicitly noted in these works, the configuration space of such larger solvent clusters is overwhelmingly huge. Therefore, it only makes sense to include them in any treatment if it is possible to focus on low-energy configurations.

The aim of this work is to start explorations along this line. We have considerable experience with global cluster-structure optimization;^[29–34] hence, we are confident of finding reliable lowest-energy structures for clusters of several dozens of molecules. In fact, in our approach, we generate low-energy solvent-cluster structures around a reaction center at several snapshots along the reaction pathway. Clearly, it would be very naive to assume that the first (and possibly the second) solvent shell around a reaction center is in a globally minimum-energy configuration at the start of a reaction and possibly also during the further course of the reaction. Indeed, this assumption is not what we propose. Instead, this approach offers an extreme view of the influence of the solvent on the reaction

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center, namely under the assumption that solvent-molecule rearrangements are much faster than progress along the reaction pathway. The advantage of this extreme view is that it is well-defined and practically realizable. A comparison of the results to those of conventional simulation approaches and to experimental data allows for conclusions in terms of how far the actual reaction dynamics in solution differ from the extreme assumption of a very quickly rearranging solvent.

We are not attempting to replace conventional simulation approaches based on MD or MC techniques for generating solvent-cluster structures. However, it is clear that these techniques explore the same, huge configuration space of solvent clusters as we do. From our search experience, we know that these spaces are enormous. Hence, it is not really certain that MD- or MC-based techniques are able to generate truly representative sets of solvent-cluster structures. Therefore, we hope that it is helpful to have another guideline in these huge configuration spaces.

S_N2 reactions have a long history of study.^[35] Because of dramatic mechanistic differences between the reactions in the gas phase and in aqueous solution, the Menshutkin reaction has been the object of extensive studies.^[36, 37, 44, 46–51] For example, Solà et al.^[37] examined the solvation effects on the Menshutkin reaction of methyl bromide with ammonia by using supermolecule and continuum models. Gao and Xia^[48] studied another Menshutkin reaction between methyl chloride and ammonia by using MC simulations with a combined quantum mechanics/molecular mechanics (QM/MM) (AM1/TIP3P) potential, in conjunction with free-energy perturbation (FEP) calculations. Truong et al.^[49] examined the same reaction by using the continuum model GCOSMO. Jay et al.^[44] studied gas-phase and aqueous trajectories of the $\text{CH}_3\text{Cl}+\text{NH}_3$ reaction with a combined atom-centered density matrix propagation (ADMP) molecular dynamics modeling and IEF-PCM approach. Amovilli et al.^[46] investigated the same reaction by using the complete active-space SCF method in conjunction with the polarizable continuum model (PCM). General agreement on the $\text{CH}_3\text{Cl}+\text{NH}_3$ reaction is that the overall reaction is endothermic by about $(110 \pm 5) \text{ kcal mol}^{-1}$ in the gas phase, thus indicating that this reaction is thermodynamically unfeasible. In solution, the barrier is lowered to about $23.5 \text{ kcal mol}^{-1}$ and the reaction is exothermic by approximately $(34 \pm 10) \text{ kcal mol}^{-1}$.

Herein, the solvent effects on the Menshutkin reaction of methyl chloride with ammonia in water are reconsidered, both by using a supermolecule model and by using a mixed explicit/implicit solvent model. As already hinted at above, we use global cluster-structure optimization methods to generate low-energy solvent clusters, thus providing an extreme view of the effect of these solvent clusters on the reaction center as an additional source of information that is complementary to conventional approaches. To the best of our knowledge, this is the first study on solvent effects that combines global cluster-structure optimization with an implicit solvent model.

The remainder of this paper is organized as follows: We first describe the computational methods in Section 2, followed by the results and discussion in Section 3. Then, we finish with some concluding remarks in Section 4.

2. Methods and Techniques

To assess the influence of the computational methods and basis sets on the structures and relative energies, various methods (such as B3LYP, B3PW91, M06-2X, MPW1K, and MP2) were employed in conjunction with the aug-cc-pVDZ or aug-cc-pVTZ basis sets to optimize the critical points along the potential-energy surface (PES), followed by single-point energy calculations with G3B3, MP2, or CCSD(T) by using the aug-cc-pVTZ basis set. For these test calculations, only the isolated $\text{CH}_3\text{Cl}+\text{NH}_3$ reaction system was considered. The calculated results are given in Tables 1, 2, and 3. Although these results exhibit some quantitative spread, they all predict similar geometries and relative energies.

Based on considerations of computational accuracy and efficiency, we chose second-order Møller–Plesset (MP2) theory in conjunction with the aug-cc-pVDZ basis set for the geometry optimization. Frequency calculations were performed at the same level of theory, to check whether the obtained species was a minimum (with all real frequencies) or a transition state

Table 1. Geometrical parameters of the reactant complex (RC), the transition state (TS), and the product complex (PC) at different levels of theory. Distances are in Å; angles are in °.

Structure	B3LYP/ aug-cc- pVDZ	B3PW91/ aug-cc- pVDZ	MPW1K/ aug-cc- pVDZ	M06-2X/ aug-cc- pVDZ	MP2/ aug-cc- pVDZ	MP2/ aug-cc- pVTZ
RC						
CN	3.379	3.502	3.365	3.170	3.216	3.302
CCI	1.816	1.799	1.782	1.798	1.804	1.786
HCN	71.72	71.18	71.28	72.35	71.56	71.45
HCCI	108.19	108.39	108.54	108.37	108.41	108.56
TS						
CN	1.774	1.780	1.821	1.817	1.829	1.804
CCI	2.450	2.488	2.404	2.424	2.418	2.400
HCN	100.69	100.03	97.66	97.88	97.52	98.13
HCCI	79.10	79.83	82.25	82.21	82.47	81.87
PC						
CN	1.593	1.571	1.533	1.549	1.555	1.549
CCI	2.749	2.740	2.768	2.774	2.780	2.741
HCN	106.81	107.44	108.32	107.99	107.72	107.89
HCCI	73.20	72.58	71.69	72.04	72.29	72.16

Table 2. G3B3 single-point energies (in kcal mol^{-1}) based on the optimized geometries in Table 1.

Species	B3LYP/ aug-cc- pVDZ	B3PW91/ aug-cc- pVDZ	MPW1K/ aug-cc- pVDZ	M06-2X/ aug-cc- pVDZ	MP2/ aug-cc- pVDZ	MP2/ aug-cc- pVTZ
R	0.0	0.0	0.0	0.0	0.0	0.0
RC	−1.5	−1.5	−1.5	−1.3	−1.4	−1.6
TS	38.4	38.5	38.6	38.5	38.5	38.5
PC	36.9	36.9	37.5	37.3	37.3	37.2
P	122.1	122.2	122.4	122.0	122.1	122.0

Table 3. Single-point energies (in kcal mol⁻¹) as calculated by using G3B3, CCSD(T), and MP2 with aug-cc-pVTZ, based on MP2/aug-cc-pVDZ-optimized geometries.

Species	With ZPE			Without ZPE	
	G3B3	CCSD(T)	MP2	MP2	Jay et al. ^[a]
R	0.0	0.0	0.0	0.0	0.0
RC	-1.4	-1.3	-1.2	-2.0	-2.1
TS	38.5	38.3	39.0	36.3	32.0
PC	37.3	37.6	37.2	32.8	30.6
P	122.1	123.2	122.6	118.0	117.0
[a] Data taken from ref. [44].					

(with only one imaginary frequency), as well as to obtain zero-point energy (ZPE) and thermodynamic corrections. To confirm that the transition state connects designated isomers, intrinsic reaction coordinate (IRC) calculations were performed. Single-point energy calculations were added at the MP2/aug-cc-pVTZ level of theory, based on the MP2/aug-cc-pVDZ-optimized geometries.

Solvation was included in the following two ways: 1) By using a supermolecule model, which featured globally optimized solvent clusters around the reacting center; and 2) by using a mixed explicit/implicit solvent model. In the first approach, 50 explicit water molecules were included. This number was sufficient to cover the reaction center with at least a monolayer of solvent molecules. It has been shown that such coverage is necessary for a quantitatively correct description of solvation.^[41] Global cluster-structure optimization^[32,34] was used to generate optimal conformations of these 50 explicit water molecules, by employing the program package OGOLEM^[33] and multiple runs per case (five for the reactant complex, transition state, and product complex). To maintain a reasonable balance between accuracy and sufficient conformational exploration of the search space, these global structure optimizations were performed at the force-field level, by using the OPLS-AA force field for the reaction center and the TIP4P water model. During global optimization of the solvent clusters, the inner reaction center was structurally fixed. Further technical details on this global optimization technique have been reported previously.^[42,43]

The cluster structures that resulted from global optimization were pooled and ranked according to their force-field energies. Then, the five "top-ranked" structures of each case were locally re-optimized and re-ranked energetically, by using a two-layer ONIOM (MP2/aug-cc-pVDZ:AMBER) scheme, in which the model system (CH₃Cl+NH₃) was treated at both the QM and MM levels, whereas the 50 explicit water molecules were only treated at the MM level. The same ONIOM scheme was also employed for the calculations of frequencies and IRCs. For refinement, additional ONIOM single-point energy calculations were performed at the MP2/aug-cc-pVTZ:AMBER level of theory. All ONIOM calculations were performed with electronic embedding (EmbedCharge option).

In the second approach (mixed explicit/implicit solvation model), local optimization for the 50 explicit water molecules in the solvent cluster was performed, whilst keeping the reac-

tion center fixed, by using an ONIOM approach, that is, at the MP2/aug-cc-pVDZ:AMBER level of theory with IEFPCM (ONIOMPCM=X,SMD). For refinement, single-point energy calculations were performed at the MP2/aug-cc-pVTZ:AMBER level of theory, with IEFPCM and SMD (ONIOMPCM=X). Clearly, none of the methods mentioned in this and in the previous paragraph account for electronic polarization of the solvent (MM region) by the reaction center (QM region). Therefore, this approach is approximate and, hence, all of the conclusions that are drawn in this article should be understood as provisional; method inaccuracies on our side, as well as in the works in which cited values were generated, can also contribute to the noted differences, which is not repeated on every occasion.

At several points, free energies are estimated. These values are solely based on the usual rigid-rotor, harmonic-oscillator models. We have not attempted to also estimate configurational entropies^[38,39] for our solvent clusters. Making such an estimate with reasonable accuracy is a hard task^[39] and we do not expect significant differences between the configurational entropies for the sets of solvent clusters that were obtained from our global structure optimizations at a given point on the reaction coordinate, because the cluster structures that we have examined and used during the course of this work did not differ significantly in their qualitative molecular arrangement. Clearly, solvent-cluster structures do change qualitatively along the reaction coordinate (to some extent); thus, configurational entropy may play a significant role in that regard. Moreover, including this entropy contribution would make our results more consistent with the use of continuum models. However, we leave this methodological improvement for future work. Furthermore, configurational entropy (as well as other entropy contributions) is fully contained in MC/MD approaches. The Gaussian 09^[40] package was used for all quantum chemistry calculations described herein.

3. Results and Discussion

3.1. In the Gas Phase

3.1.1. Isolated System

In this section, we briefly present and discuss the potential-energy surface of the Menshutkin reaction between CH₃Cl and NH₃, without any water involved. The reaction of CH₃Cl with NH₃ leads to a reactant complex (RC) and then proceeds through a transition state (TS). An ion-pair product complex (PC) was also found before the final product formed. The energy profile is shown in Figure 1. The calculated results including energy barrier, reaction energy, free-energy barrier, and reaction free energy are compared to those reported in previous studies in Table 4. As shown in Table 4, the calculated energy barriers and free-energy barriers are generally in reasonable agreement with those reported in previous studies. However, the reaction energies and reaction free energies tend to be higher than those obtained from experimental estimates and previous calculations, probably owing to the small basis sets (for example 6-31g(d) or 6-31g(d)) that were used in those

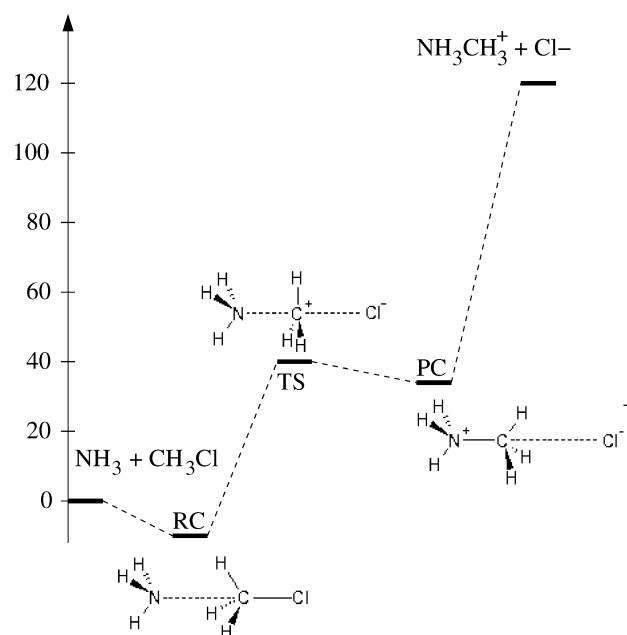


Figure 1. Energy profile (in kcal mol⁻¹) in the gas phase.

Table 4. Calculated energy barriers (ΔE), reaction energies (ΔE_p), free-energy barriers (ΔG), and reaction free energies (ΔG_p , in kcal mol ⁻¹) of the reaction $\text{CH}_3\text{Cl} + \text{NH}_3$ in the gas phase.				
Method	ΔE	ΔE_p	ΔG	ΔG_p
Su et al. ^[a]	35.1	110.6	46.5	116.9
Amovilli et al. ^[b]	38.6	114.2		
Castejon and Wiberg ^[c]	33.1			118.0
Gao and Xia ^[d]			46.7	119.0
Truong et al. ^[e]			51.4	128.3
Webb and Gordon ^[f]	33.0	106.6		
Gao ^[g] (Expt)				(110 ± 5)
this work	40.2	122.6	47.2	134.2

[a] Data taken from ref. [45]. [b] Data taken from ref. [46]. [c] Data taken from ref. [47]. [d] Data taken from ref. [48]. [e] Data taken from ref. [49]. [f] Data taken from ref. [50]. [g] Data taken from ref. [51].

studies. In addition, the relative energies (without ZPE correction) that are calculated herein are very similar to those reported by Jay et al.,^[44] as shown in Table 3.

3.1.2. Isolated System with 50 Water Molecules

In addition to the reactant complex, the transition state, and the product complex, we chose a further 12 structures along the intrinsic reaction coordinate (IRC), which was obtained in Section 3.1.1. In the following discussion, these 15 selected structures are referred to as model systems and labeled as structures n ($n=1,2,\dots,15$) along the IRC. The energy profile is shown in Figure 2 and the corresponding energies (without ZPE corrections), with structure 1 taken as zero for reference, are listed in Table 5.

As a first step in understanding the effect of solvation on the Menshutkin reaction of CH_3Cl and NH_3 , we have added 50

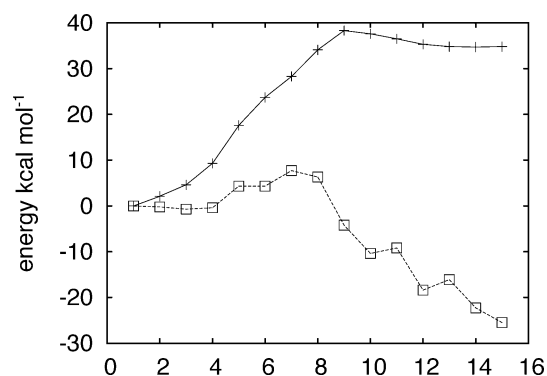


Figure 2. Energy profile (in kcal mol⁻¹) of model systems and solvated systems, without ZPE correction. Solid and dashed lines represent model and solvated systems, respectively.

Table 5. Calculated energies (without ZPE correction, in kcal mol⁻¹) of model systems and solvated systems.

Species	Isolated system	Solvated system
1	0.0	0.0
2	2.1	-0.2
3	4.6	-0.7
4	9.3	-0.4
5	17.6	4.3
6	23.7	4.3
7	28.3	7.7
8	34.1	6.3
9	38.3	-4.2
10	37.6	-10.4
11	36.5	-9.2
12	35.3	-18.4
13	34.8	-16.1
14	34.7	-22.3
15	34.8	-25.5

water molecules to each of these model systems, which are then referred to as the solvated systems. For the water molecules in these solvated systems, global structure optimization was performed as described in Section 2. For simplicity, only the first ranks of the globally optimized structures of systems 1, 7, and 15 are shown in Figure 3.

To enable a direct comparison with the gas-phase results, the relative energies (without ZPE correction), with cluster 1 taken as reference, and the corresponding energy profiles are also shown in Table 5 and Figure 2.

Figure 2 shows a considerable energy decrease for the solvated system compared to the model system; this result is further detailed in Table 5. If 50 water molecules are involved, the energy barrier is decreased to 7.7 kcal mol⁻¹, the reaction is exothermic by 25.5 kcal mol⁻¹, and the highest-energy point is shifted towards the reactants compared to the model system, consistent with Hammond's postulate^[52] and with the ADMP calculations by Jay et al.^[44] In particular, the CN and CCl bond lengths in cluster 7, which is the highest-energy point of the solvated system, are 2.17 and 2.2 Å, respectively, in agreement with the values reported by Jay et al.^[44] (2.2 and 2.13 Å). However, our calculated energy barrier of 7.7 kcal mol⁻¹ is lower than their value of 14.3 kcal mol⁻¹.

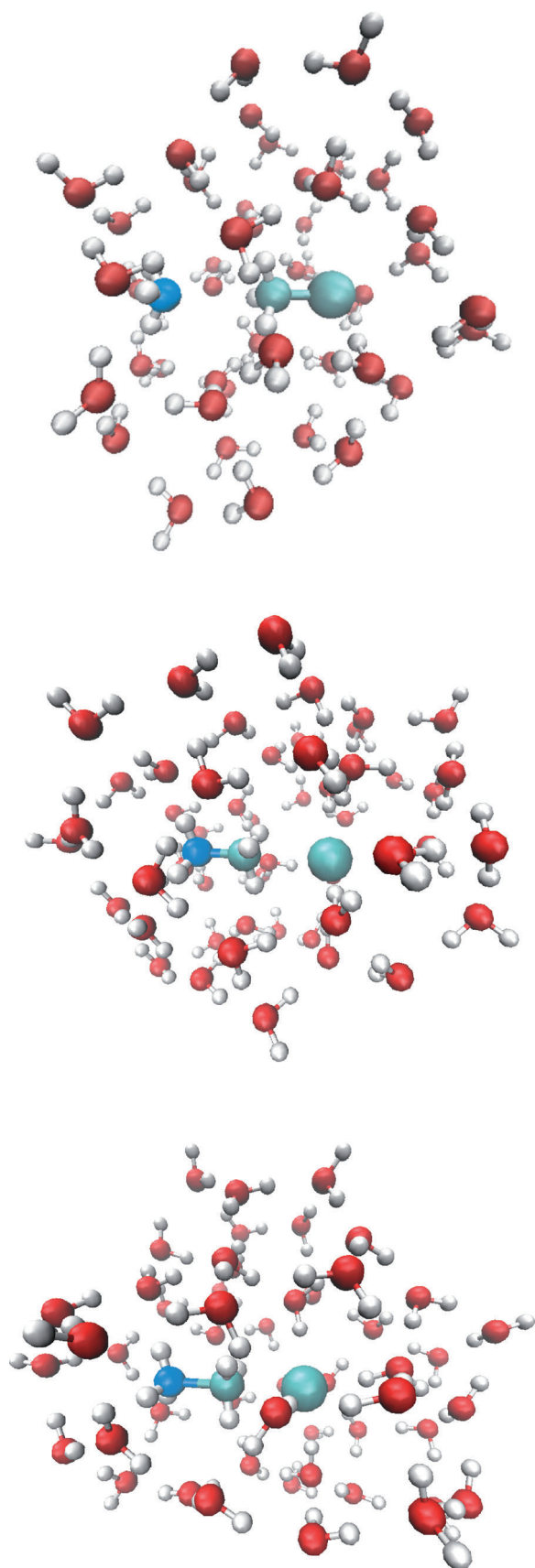


Figure 3. Globally optimized structures of clusters 1, 7, and 15.

Such a discrepancy is to be expected. After all, we have surrounded the reaction center with a rather big, explicitly treated solvation cluster and have optimized its structure anew for each of the selected structures along the IRC. This result corresponds to the assumption that the water molecules in the immediate neighborhood of the reaction center are able to instantaneously adapt to the progress of the reaction. Hence, our calculations provide a lower bound to the reaction barrier. This deviation of about 7 kcal mol^{-1} can be understood as a measure of the deviation of the actual solvation-shell behavior from instantaneous adaptability. A more-detailed analysis is presented in the next section.

3.2. In Solution

In this section, we first introduce the bulk-water effect by ONIOM single-point calculations with a continuum solvation model, based on globally optimized geometries. Then, we relax the 50 explicit water molecules in the presence of continuum solvation, whilst still keeping the model system fixed. Finally, both the model system and the 50 explicit solvating waters are relaxed in the gas phase, followed by ONIOM single-point energy calculations with continuum solvation.

3.2.1. Fixed Geometries

For the 15 solvated systems that were obtained by global optimization, ONIOM single-point energy calculations were performed with IEF-PCM and SMD. As in Section 3.1.2, for each reaction species, single-point energy calculations were performed for the first five ranked geometries. For simplicity, only the lowest energies are listed in Table 6. This table shows that, with inclusion of the bulk-solvent effect, cluster 7 is still the highest-energy point along the IRC, the energy barrier is $14.3 \text{ kcal mol}^{-1}$, and the reaction energy is $-22.0 \text{ kcal mol}^{-1}$. These values appear to be in excellent agreement with those obtained by Jay et al.^[44] (energy barrier and reaction energy of 13.5 and $-22.4 \text{ kcal mol}^{-1}$, respectively). However, as shown in

Table 6. ONIOM-PCM energies (in kcal mol^{-1}) of the 15 selected structures at fixed geometries and with relaxation of the 50 water molecules.

Species	Fixed geometries	With relaxation
1	0.0	0.0
2	2.1	2.7
3	4.0	4.6
4	4.6	5.6
5	9.3	8.4
6	11.2	9.5
7	14.3	12.0
8	8.8	5.3
9	-4.9	-10.7
10	-5.3	-6.5
11	-9.2	-11.3
12	-15.5	-16.9
13	-16.5	-19.3
14	-20.6	-22.3
15	-22.0	-25.4

the next sub-section, our results become again lower as soon as structural relaxation is taken into account.

3.2.2. Solvating Water Molecules Relaxed

We performed geometry optimizations for the 50 explicit water molecules in the presence of continuum solvation with the reaction centers kept fixed. Owing to the considerable computational expense, only the structures listed in Table 6 were optimized, to have a direct comparison with the values obtained in Section 3.2.1. Notably, the optimization of clusters 9, 12, and 14 had convergence problems; thus, we optimized lower-ranked geometries as an approximation. Subsequently, single-point energy calculations were performed, with MP2/aug-cc-pVTZ:AMBER and ONIOMPCM=X,SMD.

The obtained relative energies (referenced to optimized cluster 1) are similar to those obtained in Section 3.2.1, with a largest deviation of 5.8 kcal mol⁻¹ for cluster 9, as listed in Table 6. In addition, the calculated energy barrier (12.0 kcal mol⁻¹) and reaction energy (−25.4 kcal mol⁻¹) are in reasonable agreement with those obtained by Jay et al.^[44] Our barrier is a few kcal mol⁻¹ lower, but, as already mentioned in Section 3.1.2, this is to be expected for globally optimized solvent clusters.

3.2.3. Full Geometry Optimization

We performed full geometry optimization for clusters 1, 7, and 15 with ONIOM (MP2/aug-cc-pVDZ:AMBER) in the gas phase. Our choice of these three species was based on the energy profile shown in Table 5. Specifically, we chose clusters 1, 7, and 15 as representatives of the reactant, the transition state, and the product, respectively.

First, we examined the solvent cluster of the optimized geometries of clusters 1, 7, and 15. We found that the water clusters were very similar to those obtained in Section 3.2.2. Then, we focused on the transition state cluster 7. In this structure, the CN and CCl bond lengths are 2.247 and 2.105 Å, respectively, in good agreement with values reported in previous calculations, as shown in Table 7. Energies and free energies are summarized in Table 8, again in comparison to previous results.

The calculated energy barrier (14.3 kcal mol⁻¹) and reaction energy (−23.5 kcal mol⁻¹), without ZPE corrections, are in excellent agreement with those reported by Jay et al.^[44] On the inclusion of ZPE corrections, the agreement between the reaction energy in this work and the previous calculations be-

Table 7. Calculated geometrical parameters of the transition state, in comparison to previous results. Distances are in Å.

Bond	Jay et al. ^[a]	Gao and Xia ^[b]	Truong et al. ^[c]	Su et al. ^[d]	This work
CN	2.158	1.96	2.215	2.202	2.247
CCl	2.209	2.09	2.183	2.237	2.105

[a] Data taken from ref. [44]. [b] Data taken from ref. [48]. [c] Data taken from ref. [49]. [d] Data taken from ref. [45].

Table 8. Energy barriers (ΔE), reaction energies (ΔE_{rp}), free-energy barriers (ΔG), and reaction free energies (ΔG_{rp}) of the reaction $\text{CH}_3\text{Cl}+\text{NH}_3$; available theoretical and experimental results are also listed. All energies are in kcal mol⁻¹.

Method	ΔE	ΔE_{rp}	ΔG	ΔG_{rp}
Su et al. ^[a]	14.2	−30.1	21.8	−29.2
Amovilli et al. ^[b]			20.5	−29.1
Gao and Xia ^[c]			26.3	−18.0
Truong et al. ^[d]			24.8	−16.5
Chuang et al. ^[e]			21.6	−31.7
Gao ^[f] (Expt)			23.5	(−34±10)
ONIOM ^[g]	15.2	−18.3	18.4	−17.1
ONIOM-PCM ^[g]	20.3	−17.3	23.4	−16.1

[a] Data taken from ref. [45]. [b] Data taken from ref. [46]. [c] Data taken from ref. [48]. [d] Data taken from ref. [49]. [e] Data taken from ref. [53]. [f] Data taken from ref. [51]. [g] This work (fully relaxed stationary points of the reaction $\text{CH}_3\text{Cl}+\text{NH}_3$ in 50 water molecules).

comes worse, whereas the energy barrier is still in good agreement with the value reported by Su et al.^[45] However, even with the geometries relaxed, there are numerical differences between the free-energy barrier and the reaction free energy calculated in this work and the available results as shown in Table 8. The calculated energy profile (with ZPE correction) and free-energy profile (298.15 K, 1 atm) are shown in Figure 4.

The inclusion of bulk-solvent effects in ONIOM single-point energy calculations with continuum was also performed. The (free) energy profile is shown in Figure 4 and the corresponding energies are listed in Table 8. With the inclusion of bulk-solvent effects, our calculated free-energy barrier is in excellent agreement with previous experimental and computational results. Our calculated reaction free energy is very close to the value reported by Gao and Xia^[48] and by Truong et al.,^[49] but lower than the experimental estimate.

Again, if differences to previous results or to experiment occur, our numbers tend to be lower, consistent with globally optimized solvent-cluster structures, which yield lower limits. Or, conversely, the observed deviations are a measure of the inability of the water molecules in the solvent cluster to immediately adjust to every progress of the reaction center along the reaction coordinate.

Here and in the previous sections, we have noted several times that our results (based on globally optimized solvent clusters) tend to be in good agreement with previous (conventional) studies and that any differences are on the order of a few kcal mol⁻¹, which is small in several respects, for example, compared to the energetics of the whole reaction or compared to the energy of a hydrogen bond. This result conveys the impression that solvent dynamics are fast compared to progress along the reaction coordinate. Of course, more-detailed studies (both static and dynamic) are needed to support or disprove this tentative conclusion.

3.3. Solute Driven or Solvent Driven

The question of whether a reaction in solution is driving changes in its surrounding solvent shell(s) or, conversely, if the

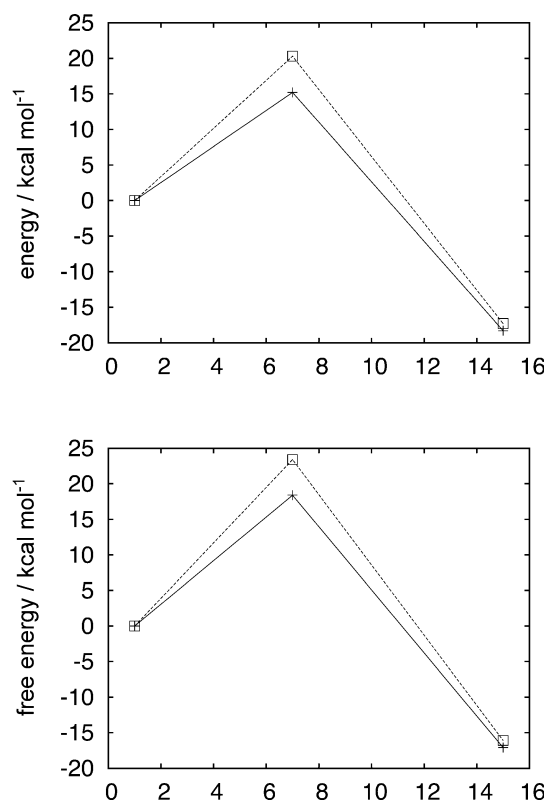


Figure 4. ZPE-corrected energies (upper panel) and free energies (lower panel) for clusters 1, 7, and 15 of the reaction $\text{NH}_3 + \text{CH}_3\text{Cl}$. Solid lines represent calculations at the ONIOM(MP2/aug-cc-pVTZ:AMBER) level of theory; dashed lines represent calculations at the ONIOM(MP2/aug-cc-pVTZ:AMBER)-PCM level of theory.

reaction progress is driven by solvent fluctuations has frequently been debated, such as in early studies of the $\text{S}_{\text{N}}2$ reaction.^[35] Also, part of this debate is whether these notions are adequate and helpful or misguided and misleading. Providing decisive support for any of these views is far beyond the scope of this article. However, we do think that it is helpful to consider the inherent timescales of reaction progress and of solvent-cluster rearrangement. In this spirit, the previous parts of this article have focused on the extreme assumption that solvent rearrangement is fast compared to reaction progress. In this section, we examine the opposite case.

To this end, we performed two kinds of calculations on our model reaction. In the first one, we placed snapshots of the reaction center into solvent clusters that were optimized for different reaction-center configurations. At the limit of slowly moving solvent molecules, the solvent cluster is actually static. A simple way of assessing the influence of such a misfit solvent cluster on the reaction center is to let the latter relax.

In several experiments of this kind, we obtained rather different and striking results. For example, we put the reactant complex (RC) in the water cluster of 7 and performed a geometry optimization of the reaction species whilst keeping the 50 solvating waters fixed. A product-like species (H_3CNH_3^+ with Cl^-) was obtained thereafter. Clearly, a transition-state-like solvent-cluster conformation facilitates the reaction so strongly that it can proceed without any barrier.

Besides the water cluster of structure 7, we also examined the water clusters of other structures, such as structures 2, 5, 8, and 9. We found that, in the water cluster of structure 9, we could also get the same product-like species ($\text{H}_3\text{CNH}_3^+ + \text{Cl}^-$). In the water cluster of structure 2, we obtained another product-like species ($\text{H}_3\text{CNH}_2 + \text{HCl}$). In water clusters 5 and 8, there was an increase in the distance between NH_3 and CH_3Cl and also a change in orientation, but, otherwise, the reaction center remained these two separate species, that is, in a reactant-like configuration.

In summary, in a regime in which water re-organization is slow compared to the progress of the reaction, certain water-cluster configurations around the reaction center have the ability to completely eliminate the reaction barrier, whereas others even deflect the reaction center away from its pre-orientation towards the reaction. Large effects owing to solvent fluctuations and their importance to the overall reactive event were already noted a long time ago, for example, by Hwang et al.,^[35] from a quite-different perspective. The new aspect here is that our solvent configurations do not arise from MD or MC sampling (in which they could be classified as “rare accidents”), but from global optimization (for various snapshots of the reaction center along its reaction path), that is, they are definitely very low-energy arrangements.

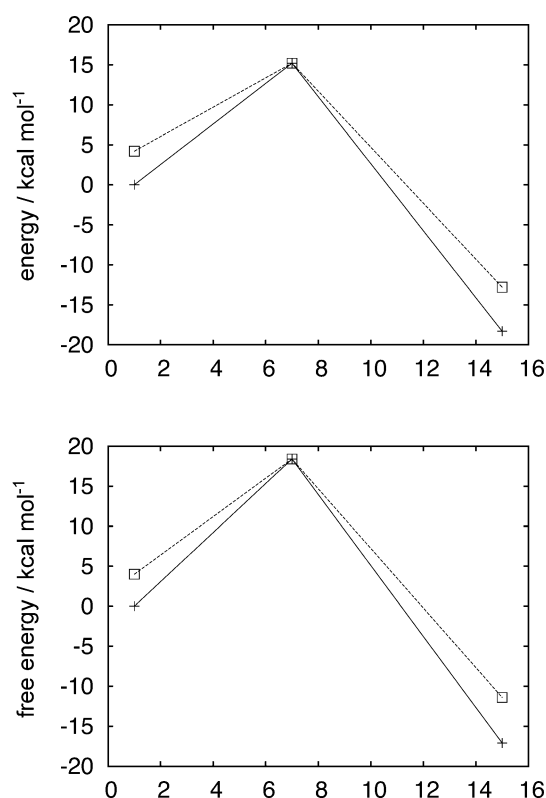
In our analysis, it may seem disturbing that there appears to be no simple pattern by which water clusters influence the reaction. As just reported, from the three cluster structures 7, 8, and 9, which are sequentially numbered along the reaction pathway, clusters 7 and 9 are reaction promoters whereas cluster 8 actually hinders the reaction (if the initial reactant complex is incorporated instead of their “proper” reaction-center structures, that is, 7, 8, or 9, respectively). The simple solution to this mystery is that the global optimizations for all of these points on the reaction pathway are completely independent from each other. Because the conformational space of a cluster of 50 water molecules is unimaginably vast, the chance of finding the best water-cluster structures around successive reaction pathway structures n and $n+1$ that bear some simple relationship to each other (such that cluster n can transform into cluster $n+1$ with little movement of a few water molecules) is zero for all practical purposes.

To re-emphasize this point from a different perspective, we performed further test calculations. We performed full optimization of the endpoints (labeled as I1 and I2) of the IRC that was obtained in Section 3.2.3. The calculated ONIOM (free) energies of I1 and I2, as well as of clusters 1, 7, and 15 (Section 3.2.3) are listed in Table 9. The (free) energy of cluster 1 was used as a reference. The corresponding (free) energy profile is shown in Figure 5.

For a proper interpretation of these data, it should be remembered that reaction-center structures 1, 7, and 15 were obtained by full geometry optimization, by using globally optimized structures 1, 7, and 15 (Section 3.1.2) as the starting structures. Then, an IRC calculation was performed, starting at the transition state, that is, cluster 7, at the MP2:AMBER level of theory, only considering the reaction center and the 50 ex-

Table 9. Energies and reaction free energies (298.15 K, 1 atm), based on the optimized geometries in the reaction $\text{CH}_3\text{Cl}+\text{NH}_3$. All energies are in kcal mol^{-1} .

Species	<i>E</i>	<i>G</i>
1	0.0	0.0
7	15.2	18.4
15	−18.3	−17.1
I1	4.2	4.0
I2	−12.8	−11.4

**Figure 5.** ZPE-corrected energies (upper panel) and free energies (lower panel) calculated at the ONIOM(MP2/aug-cc-pVTZ:AMBER) level of theory for the reaction $\text{NH}_3+\text{CH}_3\text{Cl}$. Solid lines represent clusters 1, 7, and 15; dashed lines represent endpoint I1, cluster 7, and endpoint I2.

plicit water molecules. I1 and I2 were obtained as fully relaxed endpoints of this IRC.

As shown in Table 9 and Figure 5, the (free) energies of I1 and I2 are higher than those of clusters 1 and 15, respectively. Because the geometries of the reaction center of cluster 1 and endpoint I1 are very similar, as well as of cluster 15 and endpoint I2, it can be concluded that these energy differences are largely due to re-organizations within the water clusters. However, in contrast to the first set of test calculations reported in this subsection, here, the construction of I1 and I2 through smooth IRC calculations ensures that there are comparatively simple water-structure re-organizations along this IRC, which could presumably be realized in actual dynamics. Hence, if a non-equilibrium solvent fluctuation (costing about 4–6 kcal mol^{-1}) establishes a transition-state-like solvent cluster around

the reactant complex, the effective barrier for the reaction in this cluster is lowered by this amount (in the scenario that the cluster re-arranges itself along the reaction pathway).

The results of these test calculations suggest that the reaction of CH_3Cl with NH_3 can indeed be facilitated by water-cluster re-organization into an appropriate structure for the reaction to happen; then, the reaction proceeds and, finally, the water cluster re-organizes again to stabilize the product. This picture is rather conventional, but, apparently, it is realistic, even in an explicitly treated, rather large solvent cluster. In addition, we can now attach realistic numbers to the energy lowering that accompanies such events.

Considerably less conventional is the finding that other solvent re-arrangements may actually influence the reaction so strongly that the energy profile is not only altered quantitatively, but also qualitatively. Of course, the probability of such more-extreme events happening in reality depends on the (free) energy difference that separates such an optimally pre-organized solvent cluster from a more-typical solvent cluster around the reactant complex. However, even if such a strongly favorable excursion of the solvent cluster structure is a rare event (on the timescale of water-cluster re-organizations, the elementary steps of which happen about once per picosecond),^[54–56] the reaction itself is a rare event anyway, on a similar timescale. Hence, solvent fluctuations that influence the reaction very strongly (for example, to the point of completely eliminating the barrier) may be noticeable, even if they are rare. Unfortunately, such events are difficult to capture in full dynamics simulations in explicitly treated bulk-like solvent.

4. Conclusions

We have examined the Menshutkin reaction of CH_3Cl with NH_3 in water as the solvent by using a unique approach. Large, explicitly treated solvent clusters with globally optimized structures around the reaction center, at several points along the reaction coordinate, were included in a mixed explicit/implicit solvent treatment, by employing ONIOM calculations within a continuum model.

Depending on which parts of the full model are included and how extensive local re-optimizations are applied, the resulting energetics either match or are several kcal mol^{-1} below available literature data. Qualitatively, these findings are not surprising. Re-organization of the local structure in water occurs on the picosecond timescale; hence, the re-organization can be expected to keep up with the reaction progress to a large extent; if this is not the case in reality, our numbers will be lower bounds, because this approach inherently assumes instantaneous changes in the structure of the water molecules. For this qualitative analysis, we have provided quantitative estimates in this article, in terms of energy differences, as a measure of the sluggishness of the solvent cluster. Clearly, the levels of theory that were used herein were not perfect; therefore, some of these deviations may also be owing to modeling errors. However, the numbers that we have obtained so far are consistent with the tentative conclusion that the water-cluster rearrangement is probably not significantly slower than the

progress along the reaction coordinate. In addition, this work suggests that conventional separations between the reaction center and the surrounding solvent are problematic, at least if no (or only very few) solvent molecules are treated explicitly. Larger groups of solvent molecules should be an integral part of the overall reaction coordinate.

In addition, there were also surprising findings. At least in this case, the influence of large, explicit solvent clusters on the reaction center can be far greater than just changing the reaction-energy profile to some extent. Without expecting this result and without extensive searching, we found solvent clusters that could deflect the reaction and others that could eliminate the reaction barrier completely. This result extends earlier examinations of the importance of solvent fluctuations in that all of the solvent configurations that we examined resulted from global cluster-structure optimizations and, hence, were close to optimal for certain snapshots of the reaction center along the reaction pathway. Even if solvent cluster re-arrangements are faster than the reaction progress, it is conceivable that a fluctuation in the solvent-cluster re-arrangement that favors (or disfavors) the reaction persists long enough to exert a marked influence on the reaction progress. Disappointingly, however, the conformational space of solvent clusters of this size is vast. Hence, the attainment of cluster structures of this kind can be assumed to be a rare event, which makes it difficult to observe in explicit dynamics simulations that can only capture tiny fractions of the huge numbers of possibilities. More generally and less speculatively, different solvent configurations around a reaction center may induce different reaction pathways. This possibility will be investigated in future work.

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Keywords: cluster compounds • evolutionary algorithms • quantum chemistry • solvent effects • structure optimization

- [1] C. J. Li, *Chem. Rev.* **2005**, *105*, 3095.
- [2] H. Zipse, G. Apaydin, K. N. Houk, *J. Am. Chem. Soc.* **1995**, *117*, 8608.
- [3] M. Remko, B. M. Rode, *Struct. Chem.* **2004**, *15*, 223.
- [4] X. Lin, C. Zhao, D. L. Phillips, *J. Org. Chem.* **2005**, *70*, 9279.
- [5] B. Bogdanov, T. B. McMahon, *Int. J. Mass Spectrom.* **2005**, *241*, 205.
- [6] K. Takahashi, Z. C. Kramer, V. Vaida, R. T. Skodje, *Phys. Chem. Chem. Phys.* **2007**, *9*, 3864.
- [7] W. L. Jorgensen, *Acc. Chem. Res.* **1989**, *22*, 184.
- [8] K. V. Nelson, I. Benjamin, *J. Chem. Phys.* **2009**, *130*, 194502.
- [9] Y. Shim, H. J. Kim, *J. Phys. Chem. B* **2008**, *112*, 2637.
- [10] B. L. Kormos, C. J. Cramer, *J. Org. Chem.* **2003**, *68*, 6375.
- [11] F. Ruff, Ö. Farkas, *J. Phys. Org. Chem.* **2008**, *21*, 53.
- [12] M. Seth, H. M. Senn, T. Ziegler, *J. Phys. Chem. A* **2005**, *109*, 5136.
- [13] R. Robiette, J. Richardson, V. K. Aggarwal, J. N. Harvey, *J. Am. Chem. Soc.* **2006**, *128*, 2394.
- [14] G. Alagona, C. Ghio, *Int. J. Quant. Chem.* **2010**, *110*, 765.
- [15] J. Tomasi, *Theor. Chem. Acc.* **2004**, *112*, 184.
- [16] J. Tomasi, *WIREs Comput. Mol. Sci.* **2011**, *1*, 855.
- [17] Y. Liu, B. A. Brent, A. Hengge, D. M. York, *Biochemistry* **2006**, *45*, 10043.
- [18] P. Bandyopadhyay, M. S. Gordon, *J. Chem. Phys.* **2000**, *113*, 1104.
- [19] A. H. De Vries, P. T. Van Duijnen, A. H. Juffer, J. A. C. Rullmann, J. P. Dijkman, H. Merenga, B. T. Thole, *J. Comput. Chem.* **1995**, *16*, 37.
- [20] Q. Cui, *J. Chem. Phys.* **2002**, *117*, 4720.
- [21] P. Bandyopadhyay, M. S. Gordon, B. Mennucci, J. Tomasi, *J. Chem. Phys.* **2002**, *116*, 5023.
- [22] E. F. da Silva, H. F. Svendsen, K. M. Merz, Jr., *J. Phys. Chem. A* **2009**, *113*, 6404.
- [23] G. Estiu, K. M. Merz, Jr., *J. Phys. Chem. B* **2007**, *111*, 6507.
- [24] J. M. Hayes, S. Bachrach, *J. Phys. Chem. A* **2003**, *107*, 7952.
- [25] C. O. da Silva, B. Mennucci, T. Vreven, *J. Phys. Chem. A* **2003**, *107*, 6630.
- [26] Z. Mucsi, A. Szabó, I. Hermecz, Á. Kucsman, I. G. Csizmadia, *J. Am. Chem. Soc.* **2005**, *127*, 7615.
- [27] D. Sicinska, P. Paneth, D. G. Truhlar, *J. Phys. Chem. B* **2002**, *106*, 2708.
- [28] A. Fernández-Ramos, J. A. Miller, S. J. Klippenstein, D. G. Truhlar, *Chem. Rev.* **2006**, *106*, 4518.
- [29] B. Hartke, *J. Phys. Chem.* **1993**, *97*, 9973.
- [30] S. K. Gregurick, M. H. Alexander, B. Hartke, *J. Chem. Phys.* **1996**, *104*, 2684.
- [31] B. Hartke, *J. Comput. Chem.* **1999**, *20*, 1752.
- [32] B. Hartke, *Angew. Chem.* **2002**, *114*, 1534; *Angew. Chem. Int. Ed.* **2002**, *41*, 1468.
- [33] J. M. Dieterich, B. Hartke, *Mol. Phys.* **2010**, *108*, 279.
- [34] B. Hartke, *WIREs Comput. Mol. Sci.* **2011**, *1*, 879.
- [35] J. K. Hwang, G. King, S. Creighton, A. Warshel, *J. Am. Chem. Soc.* **1988**, *110*, 5297.
- [36] P. F. Su, W. Wu, C. P. Kelly, C. J. Cramer, D. G. Truhlar, *J. Phys. Chem. A* **2008**, *112*, 12761.
- [37] M. Sola, A. Lledós, M. Duran, J. Bertrán, J. L. M. Abboud, *J. Am. Chem. Soc.* **1991**, *113*, 2873.
- [38] M. Karplus, J. N. Kushick, *Macromolecules* **1981**, *14*, 325.
- [39] P. H. Nguyen, P. Derreumaux, *Phys. Chem. Chem. Phys.* **2012**, *14*, 877.
- [40] Gaussian 09, Revision A.2, M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H. P. Hratchian, A. F. Izmaylov, J. Bloino, G. Zheng, J. L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J. A. Montgomery, Jr., J. E. Peralta, F. Ogliaro, M. Bearpark, J. J. Heyd, E. Brothers, K. N. Kudin, V. N. Staroverov, R. Kobayashi, J. Normand, K. Raghavachari, A. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, N. Rega, J. M. Millam, M. Klene, J. E. Knox, J. B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R. E. Stratmann, O. Yazyev, A. J. Austin, R. Cammi, C. Pomelli, J. W. Ochterski, R. L. Martin, K. Morokuma, V. G. Zakrzewski, G. A. Voth, P. Salvador, J. J. Dannenberg, S. Dapprich, A. D. Daniels, Ö. Farkas, J. B. Foresman, J. V. Ortiz, J. Ciołowski, D. J. Fox, Gaussian, Inc., Wallingford CT, **2009**.
- [41] J. Thar, S. Zahn, B. Kirchner, *J. Phys. Chem. B* **2008**, *112*, 1456.
- [42] S. Frick, diploma thesis, University of Kiel, **2011**.
- [43] S. Frick, B. Hartke, *unpublished results*.
- [44] A. N. Jay, K. A. Daniel, E. V. Patterson, *J. Chem. Theory Comput.* **2007**, *3*, 336.
- [45] P. Su, F. Ying, W. Wu, P. C. Hiberty, S. Shaik, *ChemPhysChem* **2007**, *8*, 2603.
- [46] C. Amovilli, B. Mennucci, F. M. Floris, *J. Phys. Chem. B* **1998**, *102*, 3023.
- [47] H. Castejon, K. B. Wiberg, *J. Am. Chem. Soc.* **1999**, *121*, 2139.
- [48] J. Gao, X. Xia, *J. Am. Chem. Soc.* **1993**, *115*, 9667.
- [49] T. N. Truong, T. T. Truong, E. V. Stefanovich, *J. Chem. Phys.* **1997**, *107*, 1881.
- [50] S. P. Webb, M. S. Gordon, *J. Phys. Chem. A* **1999**, *103*, 1265.
- [51] J. Gao, *J. Am. Chem. Soc.* **1991**, *113*, 7796.
- [52] G. S. Hammond, *J. Am. Chem. Soc.* **1955**, *77*, 334.
- [53] Y. Y. Chuang, C. J. Cramer, D. G. Truhlar, *Int. J. Quantum. Chem.* **1998**, *70*, 887.
- [54] S. Ashihara, N. Huse, A. Espagne, E. T. J. Nibbering, T. Elsaesser, *J. Phys. Chem. A* **2007**, *111*, 743.
- [55] K. A. Tay, F. Bresme, *Phys. Chem. Chem. Phys.* **2009**, *11*, 409.
- [56] K. Hyeon-Deuk, K. Ando, *J. Chem. Phys.* **2010**, *132*, 164507.

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